1. ScamShield US

- Agentic RAG for consumer fraud / identity-theft education
- Stack constraint: AWS EC2 + S3 only • open-source models • boto3
- Classify → Retrieve top-5 → Answer with local Qwen

2. Problem & goals

- Help consumers interpret suspicious SMS/messages safely
- Compare Traditional, Advanced (DistilBERT), and Hybrid classifiers
- Ground answers in official FTC/CISA/SSA/… guidance

3. System architecture

- Angular UI (:4200) → FastAPI/Uvicorn (:8000)
- Agents: fraud detection → retrieval → Qwen answer → validation
- S3DataStore (boto3) for knowledge, models, indexes, metrics

4. Offline pipeline

- seed knowledge → prepare splits → train models → build index
- evaluate approaches → build dashboard viz
- All durable outputs written to the course S3 prefix

5. Online investigation workflow

- Hybrid classifier scores the message
- Hybrid BM25 + FAISS returns 5 evidence chunks
- Qwen2.5-1.5B decoder generates a cited educational answer

6. Models from the course concept list

- Classical: TF-IDF + Logistic Regression / Linear SVM
- Pretrained Transformers: DistilBERT, MiniLM, Qwen (decoder attention)
- Hybrid cascade margin = 0.05 (validation-chosen)

7. Results (held-out test)

- Hybrid macro-F1 0.983 • accuracy 0.992 • ~1.1% routed to DistilBERT
- Traditional and Advanced are strong; Hybrid wins on macro-F1
- Primary metric = macro-F1 (rare subtypes matter)

8. Demo & reproducibility

- Code/ + uv sync + export AWS credentials
- uvicorn backend.app.main:app --port 8000 • npm start in frontend/
- Educational use only — not legal advice